The `dihiggs` model

A physics-structured neural network for the HEFT di-Higgs mass spectrum

October 2026

Abstract

This note explains, step by step, the model implemented in `dihiggs/heftnet.py`. The goal is to predict the gluon-fusion di-Higgs cross section in each bin of the di-Higgs invariant mass m_{hh} as a function of m_{hh} and of five Wilson coefficients of the Higgs effective field theory (HEFT). The central idea is to *not* ask the neural network to learn what physics already tells us: the dependence on the Wilson coefficients is a known polynomial with 23 terms. The network learns only the 23 unknown functions of m_{hh} that multiply those terms. We first derive the polynomial, then describe the network, the data, the training, and the testing, and end with some points worth noticing.

Contents

1 What we want to model, and why **2**

2 The physics: why σ is a polynomial in θ **3**
2.1 The amplitude is linear in the couplings . . . 3
2.2 Squaring the amplitude gives 15 terms . . . 3
2.3 Next-to-leading order adds 8 more terms . . . 3

3 The model **4**
3.1 Architecture . . . 4
3.2 What the structure buys us . . . 5

4 The data **6**

5 Training **7**

6 Testing **7**

7 Points worth noticing **7**

A The 23 monomials **8**

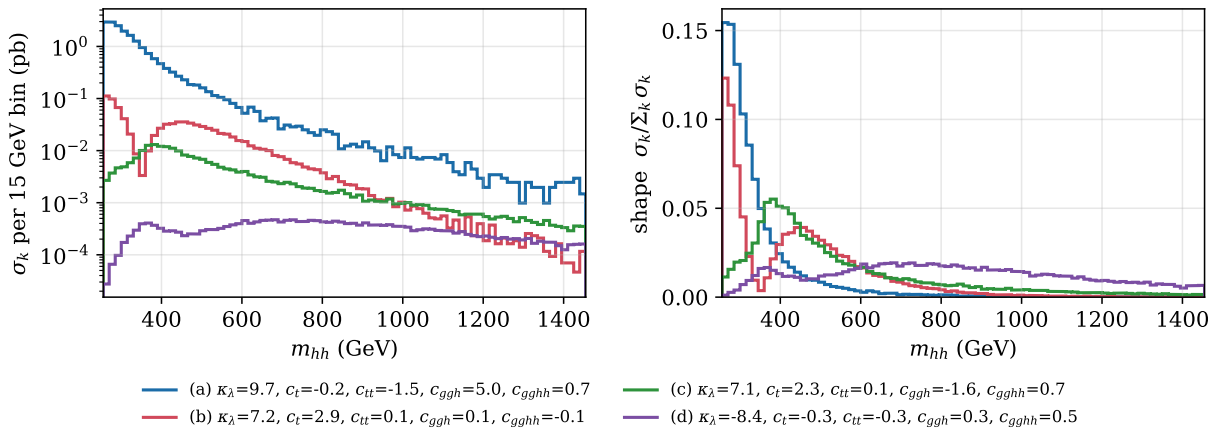


Figure 1: Four POWHEG spectra from the held-out test set, chosen to show the range of shapes. Left: cross section per 15 GeV bin (log scale). Right: the same spectra divided by their totals, so that only the shapes are compared.

1 What we want to model, and why

In gluon fusion, $gg \to hh$, new physics described by HEFT changes both the size and the shape of the di-Higgs invariant-mass spectrum. The relevant part of the HEFT Lagrangian is [1]

$$\mathcal{L}_{\text{HEFT}} \supset -m_t\Big(c_t\,\frac{h}{v} + c_{tt}\,\frac{h^2}{v^2}\Big)\bar{t}t - \kappa_\lambda\,\frac{m_h^2}{2v}\,h^3 + \frac{\alpha_s}{8\pi}\Big(c_{ggh}\,\frac{h}{v} + c_{gghh}\,\frac{h^2}{v^2}\Big)G^a_{\mu\nu}G^{a,\mu\nu}, \tag{1}$$

where $v \approx 246$ GeV. Each coefficient has a simple meaning:

coefficient	interaction it controls	SM value
$\kappa_\lambda \equiv c_{hhh}$	Higgs self-coupling hhh	1
c_t	top Yukawa coupling $t\bar{t}h$	1
c_{tt}	contact coupling $t\bar{t}hh$	0
c_{ggh}	effective (point-like) coupling ggh	0
c_{gghh}	effective (point-like) coupling $gghh$	0

We collect them into $\theta = (\kappa_\lambda, c_t, c_{tt}, c_{ggh}, c_{gghh})$.

We want a function

$$\sigma(m_{hh}, \theta) = \text{cross section in a 15 GeV bin centred at } m_{hh}, \tag{2}$$

valid everywhere in a broad region of the 5-dimensional θ space. Such a function lets us *reweight* existing simulated events to any other θ, instead of running a new simulation for every point, and it is an ingredient of the simulation-based construction of confidence sets in θ that motivates this project [2, 3].

Figure 1 shows four of the simulated spectra. Notice how different they are: their sizes span three orders of magnitude, some fall steeply from threshold, one peaks near 700 GeV, and spectrum (b) has a sharp dip near 330 GeV, a sign of destructive interference between different contributions to the amplitude. The model has to describe all of these with one set of parameters.

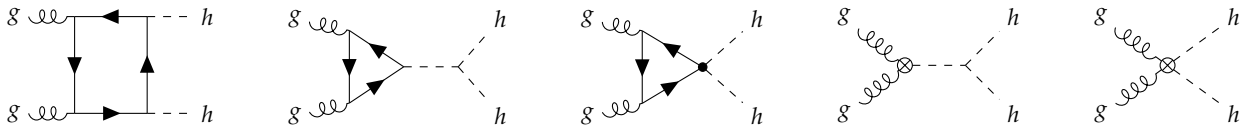


Figure 2: The five classes of LO diagrams for $gg \to hh$ in HEFT and the coupling combination each one carries. Solid lines are top quarks, dashed lines Higgs bosons; the dot is the $t\bar{t}hh$ contact vertex and the crossed dots are the point-like gluon–Higgs vertices of eq. (1).

2 The physics: why σ is a polynomial in θ

2.1 The amplitude is linear in the couplings

At leading order (LO) there are five classes of Feynman diagrams for $gg \to hh$ in HEFT (fig. 2). Each class contains exactly one combination of couplings: two top Yukawa vertices in the box, a Yukawa vertex and the self-coupling in the top-loop triangle, and so on. Therefore the full amplitude is a *linear* combination of five fixed sub-amplitudes,

$$\mathcal{M} = c_t^2\,\mathcal{M}_1 + c_t\kappa_\lambda\,\mathcal{M}_2 + c_{tt}\,\mathcal{M}_3 + c_{ggh}\kappa_\lambda\,\mathcal{M}_4 + c_{gghh}\,\mathcal{M}_5 \;\equiv\; \sum_{i=1}^{5} v_i(\theta)\,\mathcal{M}_i, \tag{3}$$

with

$$\boldsymbol{v}(\theta) = \big(c_t^2,\; c_t\kappa_\lambda,\; c_{tt},\; c_{ggh}\kappa_\lambda,\; c_{gghh}\big). \tag{4}$$

The sub-amplitudes $\mathcal{M}_i$ depend on the kinematics (on m_{hh} and the scattering angle) but *not* on θ. All of the θ dependence is in the five numbers v_i.

2.2 Squaring the amplitude gives 15 terms

A cross section is proportional to $|\mathcal{M}|^2$. Squaring eq. (3) gives

$$|\mathcal{M}|^2 = \sum_{i=1}^{5}\sum_{j=1}^{5} v_i\,v_j\,\mathrm{Re}\big(\mathcal{M}_i\mathcal{M}_j^*\big). \tag{5}$$

Now integrate over everything except m_{hh} (the scattering angle, and the parton distributions), and over one 15 GeV bin. Because the v_i do not depend on the kinematics, they come out of the integral:

$$\sigma(m_{hh},\theta) = \sum_{i\le j} v_i(\theta)\,v_j(\theta)\,K_{ij}(m_{hh}), \qquad K_{ij}(m_{hh}) \propto \int \mathrm{Re}\big(\mathcal{M}_i\mathcal{M}_j^*\big)\,d(\text{everything but } m_{hh}). \tag{6}$$

K is a symmetric 5×5 matrix of functions of m_{hh}, so it has $5\cdot 6/2 = 15$ independent entries. The 15 products v_iv_j, written out in the couplings, are the first 15 monomials in table 2. The diagonal entries K_{ii} are squared magnitudes and so are non-negative; the off-diagonal entries are *interference* terms and can have either sign. Negative interference is what produces dips such as the one in spectrum (b) of fig. 1.

2.3 Next-to-leading order adds 8 more terms

The training data were generated with POWHEG at next-to-leading order (NLO) in QCD. At NLO new diagram classes appear, and the polynomial grows from 15 to 23 terms [1]. One way to see

where the 8 extra terms come from is to consider the NLO diagrams in which each Higgs boson attaches to the gluons through its own single-Higgs vertex, either a top loop ($\propto c_t$) or the point-like ggh vertex ($\propto c_{ggh}$). Their coupling factors are $\boldsymbol{w} = (c_t^2,\ c_t c_{ggh},\ c_{ggh}^2)$. Interfering these with the LO amplitude gives products $w_a v_i$. The products with $w_1 = c_t^2$ reproduce LO monomials, but those with $c_t c_{ggh}$ and c_{ggh}^2 give exactly the 8 new monomials 16–23 of table 2.

Key idea

For *every* m_{hh} the cross section is a fixed polynomial in θ:

$$\sigma(m_{hh}, \theta) = \sum_{i=1}^{23} c_i(\theta)\, a_i(m_{hh}) = \boldsymbol{c}(\theta) \cdot \boldsymbol{a}(m_{hh}) \tag{7}$$

The 23 monomials $c_i(\theta)$ are *known exactly* (table 2). Only the 23 functions $a_i(m_{hh})$ are unknown. They are the integrated (and NLO-corrected) interference kernels K_{ij}. The coupling dependence and the kinematic dependence *factorise*.

The LHC Higgs Working Group determines the a_i (called dA_i there) one bin at a time with a least-squares fit [1]. The `dihiggs` model instead represents them as smooth functions of m_{hh} given by a neural network.

3 The model

3.1 Architecture

Figure 3 shows the whole model. It has two branches that meet in a dot product.

The physics branch, $\boldsymbol{c}(\theta)$. The five Wilson coefficients are turned into the 23-component vector of monomials in table 2. This branch has no trainable parameters; it is simply eq. (6) and its NLO extension written as code (`torch.column_stack` in `HEFTNet.forward`).

The learned branch, $\boldsymbol{a}(m_{hh})$. The mass alone determines the 23 functions, through

$$a_i(m_{hh}) = P_i(m_{hh})\, e^{\,Q_i(m_{hh})}, \qquad i = 1, \ldots, 23, \tag{8}$$

where

- P is a fully connected network with one input and 23 outputs: a $1 \to 25$ layer, five $25 \to 25$ layers, each followed by a ReLU, and a final $25 \to 23$ layer followed by tanh. This is 3 898 parameters.
- Q is a single linear layer with one input and 23 outputs, $Q_i(m_{hh}) = w_i\, m_{hh} + b_i$, which is 46 parameters.

Why the product $P\, e^{Q}$?

Because Q_i is linear, $e^{Q_i(m_{hh})} = e^{b_i}\, e^{w_i m_{hh}}$. The two factors have separate jobs, and P_i has a third:

- e^{b_i} sets the *overall size* of a_i. The 23 functions can differ in magnitude by orders of

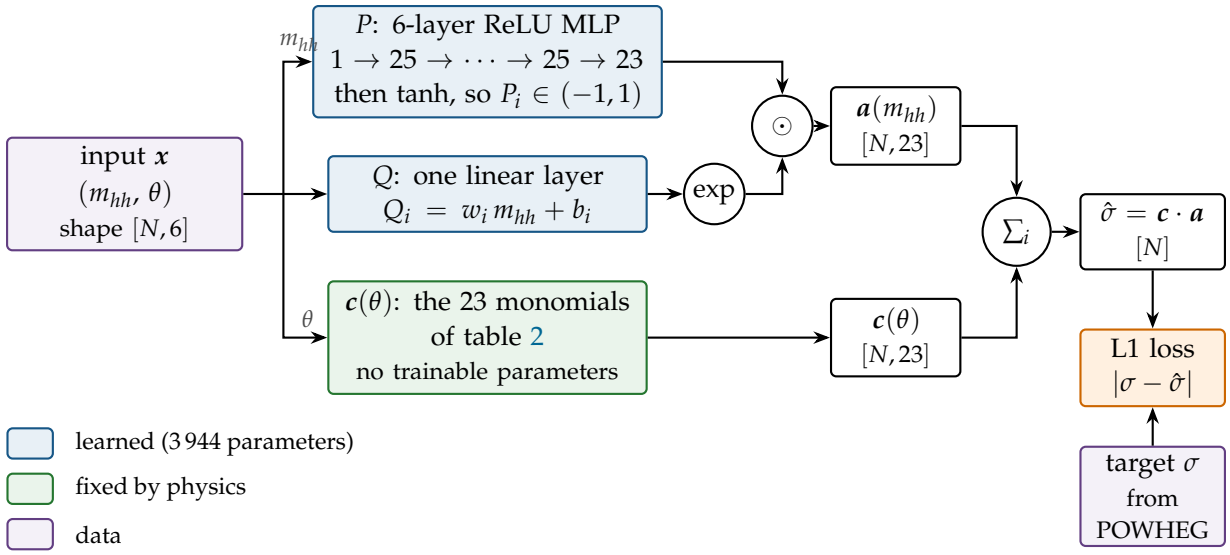


Figure 3: The `HEFTNet` model. Each input row, $\boldsymbol{x} = (m_{hh}, \theta)$ with $\theta = (\kappa_\lambda, c_t, c_{tt}, c_{ggh}, c_{gghh})$, is split. The mass m_{hh} feeds two small networks, P and Q, which give the 23 functions $a_i(m_{hh}) = P_i(m_{hh})\, e^{Q_i(m_{hh})}$. The Wilson coefficients θ feed the 23 known monomials $c_i(\theta)$. The prediction is the dot product $\boldsymbol{c} \cdot \boldsymbol{a}$, which is compared with the POWHEG cross section using the absolute (L1) loss. Blue boxes are learned; the green box is fixed by physics.

magnitude. Learning b_i, a logarithm of the size, is far easier for gradient descent than learning a tiny or huge number directly.

- $e^{w_i m_{hh}}$ supplies an *exponential fall-off* with m_{hh}, which is the gross behaviour of the spectra in fig. 1.
- $P_i \in (-1, 1)$ supplies the *shape and the sign*. A sign is needed because the interference functions can be negative. The tanh keeps P_i bounded, so the magnitude is always controlled by e^{Q_i}.

In short, e^Q is a simple learned envelope and P is a learned, bounded correction to it.

The prediction. For an input row $\boldsymbol{x} = (m_{hh}, \kappa_\lambda, c_t, c_{tt}, c_{ggh}, c_{gghh})$ the output is

$$\hat{\sigma}(\boldsymbol{x}) = \sum_{i=1}^{23} c_i(\theta)\, P_i(m_{hh})\, e^{Q_i(m_{hh})}. \tag{9}$$

Table 1 follows the tensors through `HEFTNet.forward` for a batch of N rows.

3.2 What the structure buys us

1. **Exact dependence on the couplings.** For any θ, including values outside the training region, the prediction has exactly the right polynomial form. Only the a_i are approximated. A generic network with all six inputs would have to discover this polynomial from data and could not extrapolate it.

step in `forward`	tensor	shape
input batch	$\boldsymbol{x}$	$[N, 6]$
split columns	$m_{hh}, \kappa_\lambda, c_t, c_{tt}, c_{ggh}, c_{gghh}$	six of $[N]$
monomials (`column_stack`)	$\boldsymbol{c}(\theta)$	$[N, 23]$
reshape mass	m_{hh}	$[N, 1]$
networks	$P(m_{hh})$, $Q(m_{hh})$	$[N, 23]$ each
coefficients (`coeffs`)	$\boldsymbol{a} = P \odot e^{Q}$	$[N, 23]$
row-wise dot product	$\hat{\sigma} = \sum_i c_i a_i$	$[N]$

Table 1: Tensor shapes in `HEFTNet.forward`.

2. **A one-dimensional learning problem.** The network sees only m_{hh}. Learning 23 smooth functions of one variable is much easier than learning one function of six variables.

3. **Interpretable outputs.** The $a_i(m_{hh})$ are physical quantities, namely the interference kernels, and can be plotted and compared with the binned dA_i of Ref. [1]. `heft_test.ipynb` plots them.

4. **A continuous function of m_{hh}.** The model can be evaluated at any m_{hh}, not just at bin centres.

4 The data

Spectra. The file `data/powheg_total_gauss_params_13TeV.csv.gz` contains 4 999 POWHEG NLO spectra at $\sqrt{s} = 13$ TeV. Each row holds one parameter point θ and the cross section in pb in 100 bins of 15 GeV. The parameter points are scattered roughly as Gaussians centred on the SM point, with standard deviations of about 5 for κ_λ and about 2 for the other four coefficients. Hence the name "wide Gauss".

Bins and the mass variable. Bins 0–15 are empty, and bin 16 straddles the threshold $m_{hh} = 2m_h = 250$ GeV: the POWHEG histograms start at 0 GeV with 15 GeV bins, so bin k covers $[15k, 15(k+1))$ GeV. `heft_prepare_data.ipynb` keeps bins 17–96, that is $255 \le m_{hh} < 1455$ GeV, and maps them to a dimensionless variable on the unit interval with bins of width 0.01:

$$x = \frac{m_{hh} - 255\ \text{GeV}}{1500\ \text{GeV}}, \qquad x_k = \frac{k - 17 + \frac{1}{2}}{100}, \quad k = 17, \ldots, 96. \tag{10}$$

In the code, the column called `mhh` holds this x, which runs from 0.005 to 0.795. Keeping the network input of order one is good practice.

From spectra to training rows. After shuffling the spectra:

- the first 500 are set aside, untouched, as the test set (`heft_wide_gauss_testdata_13TeV.csv`);
- each of the remaining 4 499 spectra is "diced" into 80 rows (θ, x_k, σ_k), one per bin. Rows with $\sigma_k = 0$ are dropped and the rows are shuffled, leaving 357 958 rows (`heft_wide_gauss_traindata_13TeV.csv.gz`).

Dicing is what makes the problem a standard regression: every row is one example, input $\boldsymbol{x} = (x_k, \theta)$ and target σ_k.

5 Training

Splits. `heft_train.ipynb` splits the 357 958 rows into 301 056 for training (= 147 batches of 2 048), 6 902 for validation, and 50 000 for a quick test. A random subset of the training rows, of the same size as the validation set, is used to compare training and validation losses, which is a check for overfitting.

Loss. The parameters ω of P and Q are found by minimising the empirical risk with the absolute (L1) loss,

$$R(\omega) = \frac{1}{M} \sum_{m=1}^{M} \left| \, \sigma_m - \boldsymbol{c}(\theta_m) \cdot \boldsymbol{a}(x_m; \omega) \, \right|. \tag{11}$$

The minimiser of the expected absolute loss is the conditional *median* of the target given the input. With the squared loss it would be the conditional mean. For simulated spectra the target is a deterministic function of the input up to Monte Carlo noise, so both would give essentially the same function. The L1 loss is less sensitive to occasional large fluctuations.

Optimisation. Adam with mini-batches of 2 048 rows for 200 000 iterations, about 1 360 passes through the training set. The learning rate starts at 2×10^{-3} and is halved every 50 000 iterations, giving four steps: 2×10^{-3}, 10^{-3}, 5×10^{-4}, 2.5×10^{-4}. Every 100 iterations the average losses on the training subset and the validation set are recorded. The parameters are saved whenever the validation loss drops by more than 1.5% below its best value so far.

6 Testing

`heft_test.ipynb` reloads the saved parameters and does two things.

1. It plots the 23 learned functions $a_i(m_{hh})$, the quantities the network was built to find.
2. For each of the 500 held-out spectra it evaluates $\hat{\sigma}$ at the 80 bin centres and overlays the result on the POWHEG histogram. None of these parameter points was used in training. The test is therefore a direct check of how well the model *interpolates in* θ, which is the property needed for reweighting.

7 Points worth noticing

- **Positivity is not enforced.** Nothing forces $\hat{\sigma} \geq 0$, nor the diagonal kernels to be non-negative. A well-trained model should be very close to non-negative where there are data, but it is worth checking near the dips.
- **The absolute loss weights rows by their size.** The targets span roughly 10^{-6} to 30 pb per bin. An absolute error of 10^{-4} pb is negligible for a large spectrum but a large *relative* error for a small one. Training is therefore dominated by the larger cross sections. A relative or logarithmic loss would weight the spectra more evenly, at the price of different statistical properties.
- **Identifiability.** At a fixed m_{hh}, σ is linear in the 23 unknowns a_i. They are determined uniquely only if the 23 monomials evaluated at the training points are linearly independent.

With thousands of random θ points spread widely, as here, they are. With a few points close to the SM, they would not be.

A The 23 monomials

i	$c_i(\theta)$	origin	i	$c_i(\theta)$	origin
1	c_t^4	$v_1 v_1$	13	$\kappa_\lambda^2 c_{ggh} c_t$	$v_2 v_4$
2	c_{tt}^2	$v_3 v_3$	14	$c_{gghh} c_t \kappa_\lambda$	$v_2 v_5$
3	$c_t^2 \kappa_\lambda^2$	$v_2 v_2$	15	$c_{ggh} c_{gghh} \kappa_\lambda$	$v_4 v_5$
4	$c_{ggh}^2 \kappa_\lambda^2$	$v_4 v_4$	16	$c_t^3 c_{ggh}$	$(c_t c_{ggh})\, v_1$
5	c_{gghh}^2	$v_5 v_5$	17	$c_t c_{tt} c_{ggh}$	$(c_t c_{ggh})\, v_3$
6	$c_{tt} c_t^2$	$v_1 v_3$	18	$c_t c_{ggh}^2 \kappa_\lambda$	$(c_t c_{ggh})\, v_4$
7	$\kappa_\lambda c_t^3$	$v_1 v_2$	19	$c_t c_{ggh} c_{gghh}$	$(c_t c_{ggh})\, v_5$
8	$c_t \kappa_\lambda c_{tt}$	$v_2 v_3$	20	$c_t^2 c_{ggh}^2$	$c_{ggh}^2\, v_1$
9	$c_{ggh} \kappa_\lambda c_{tt}$	$v_3 v_4$	21	$c_{tt} c_{ggh}^2$	$c_{ggh}^2\, v_3$
10	$c_{tt} c_{gghh}$	$v_3 v_5$	22	$c_{ggh}^3 \kappa_\lambda$	$c_{ggh}^2\, v_4$
11	$c_{ggh} \kappa_\lambda c_t^2$	$v_1 v_4$	23	$c_{ggh}^2 c_{gghh}$	$c_{ggh}^2\, v_5$
12	$c_{gghh} c_t^2$	$v_1 v_5$			

Table 2: The monomials in the order used in `heftnet.py`, which is also the order of Eq. (4.1) of Ref. [1]. Terms 1–15 are the products $v_i v_j$ of the LO coupling factors in eq. (4) and suffice at LO. Terms 16–23 appear at NLO. Their "origin" column shows them as an NLO coupling factor ($c_t c_{ggh}$ or c_{ggh}^2) times an LO factor v_i, as described in section 2. In the code, `klambda` $= \kappa_\lambda$, `CT` $= c_t$, `CTT` $= c_{tt}$, `CGGH` $= c_{ggh}$, `CGGHH` $= c_{gghh}$.

References

[1] L. Alasfar *et al.*, *Effective Field Theory descriptions of Higgs boson pair production*, LHC Higgs Working Group public note, arXiv:2304.01968 (included in this repository as `docs/theory.pdf`).

[2] N. Dalmasso, L. Masserano, D. Zhao, R. Izbicki and A. B. Lee, *Likelihood-free frequentist inference: bridging classical statistics and machine learning for reliable simulator-based inference*, arXiv:2107.03920.

[3] A. Al Kadhim *et al.*, *Amortized simulation-based frequentist inference for tractable and intractable likelihoods*, Mach. Learn.: Sci. Technol. (2024), doi:10.1088/2632-2153/ad218e.